Wood Wide Webs

James Bridle. "Wood Wide Webs" (excerpt). *Ways of Being: Animals, Plants, Machines: The Search for a Planetary Intelligence*. New York: Picador, 2022. 59–78.

James Bridle is a writer, artist, technologist, and and technology critic. His book *Ways of Being* questions the idea that "intelligence" belongs only to humans and to brains. He has been working in the field of artificial intelligence for many years. He is also interested in the intelligence of animals and plants. His fascinating book is full of scientific insights into the stunning diversity of natural and artificial intelligence. This chaper explores how plants hear, smell, remember, learn, and communicate.

ASK YOURSELF THESE QUESTIONS ABOUT THE READING

—What is intelligence?

—How do you engage with non-human intelligence (animals, pets, plants, AI, etc?)

—How do you experience and develop your own intelligence? Does thinking happen inside or outside your head? Is your body intelligent?

—What influences and interconnections in your work trace back to unexpected sources (from your family history, personal history, experiences with travel and nature, friendships, art, games, etc)?

—"The earliest plants were mere agglomerations of cellular tissue, lacking roots or leaves or any of the specialized structures we recognize in plants today. They were the descendants of simple marine algae which washed ashore and found some purchase on beaches and cliffs, sustaining themselves through photosynthesis alone. But around 400 million years ago...these proto-plants began to associate with fungi: to evolve lobes and fleshy organs to house mycorrhizal partners. This is the origin of all plant roots: questing limbs in search, not of food itself, but of partners in the process of producing life." (78)

—It is amazing to picture simple cells interacting with other life forms to create leaves, stems, roots and other "fleshy organs." Picture a letterform or character interacting with other organisms (such as Latin, Chinese, Korean, adn Indic characters feeding one another). How can form interact with other forms? How can this process enliven your work?

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At the beginning of 2018 I was invited to give a talk at a big, glossy event in Vancouver. The topic of the talk was to be some of the darker aspects of the internet, and particularly the effects of online video and algorithms on children: a grim subject which I didn't relish delving into. On the other hand, I'd never been to the west coast of Canada before, and the trip promised encounters with interesting new people and landscapes.

As a speaker at the conference, I was offered a range of regional 'experiences', most of which I missed out on as I hid in my hotel room, frantically rewriting and rehearsing my talk. But there was one that seemed too good to miss: a tour of the redwood forests on the city's edge, led by a biologist from the University of British Columbia. So I climbed aboard a bus with a couple of dozen other attendees and went for a walk in the forest.

Our guide was Professor Suzanne Simard, who has spent decades studying the giant redwood forests of Canada's Pacific coast. As we walked among the mossy trunks, she explained how the huge trees were intimately connected to one another – and to places far outside the forest itself. For example, a significant proportion of the essential nitrogen that the trees and other forest plants take up through their roots is ultimately derived from very far away indeed – from the middle of the Pacific Ocean, thousands of miles distant.

We know this because there's a particularly heavy nitrogen isotope called nitrogen-15 which is much more common in marine algae than in most terrestrial vegetation. Yet it's found in surprising amounts in coastal forests, and the way it gets there is quite marvellous. The fish which feed on algae and phytoplankton in the deep ocean become

enriched with nitrogen-15. One of these fish is the Pacific salmon, which every year returns to spawn in the same rivers and streams from which they hatched, on the coast of the western United States and Canada. As they swim upstream, they're caught and eaten by the black and grizzly bears who have made their own journeys from far inland to congregate along the salmon rivers during the spawning season. The remains of the fish – as well as the bears' copious droppings – fertilize the trees and other forest plants and mark them with that particular form of nitrogen, ferried over thousands of miles in the bodies of other species.[1]

Similar networks of distribution, and even of mutual aid, exist beneath the forest floor, as Professor Simard explained. The trees can't access much of this nitrogen directly: instead, they rely on specialized fungi growing on, around, and even through their roots. These fungi absorb and process the nitrogen with their own enzymes, and pass the resulting nutrients onto the trees. The dense networks of interconnected fibres through which this exchange takes place are called mycorrhiza – a literal growing-together of the Greek words for fungus (*mykós*) and root (*riza*) – and we're only just beginning to understand their significance.

The mycorrhizal trade is not a one-sided deal. In turn, the trees supply the fungi with the sugar and carbon they produce from photosynthesis: solar products for underground dwellers. Even more remarkably, the trees connect to one another through this network. Approaching one of the largest trees along the path, Professor Simard introduced us to it as a 'mother tree', which serves as a central hub in the mycorrhizal network. As new seedlings emerge from the undergrowth, the mother tree infects them with the fungi, and uses it to supply them with the nutrients they need to grow.

The trees are selective in whom they choose to help: mother trees favour their own offspring, sending more carbon to them than to unrelated seedlings, and shifting their roots to make room for them to grow. But they're not selfish, and will go to the aid of other species in times of need. In the summer, the shaded fir trees receive carbon and sugar from the taller birch trees, while in autumn the firs return the favour as the birches start to lose their leaves.

This living network is not just about sharing food; it's also about

sharing information. If one tree in the forest is attacked by insects, it releases warning chemicals into the network. The other trees pick up the signal and respond by producing their own chemical defences. Standing among the giant trunks, listening to Simard and her colleagues speak, I began to realize that the forest was filled with a constant hum of unseen signs and unheard chatter. Decisions were being made, agreements reached, bargains made and broken. The trees were speaking to one another. A whole world of voices I had never heard – whose existence I had never even suspected – burst into my consciousness.

As mesmerized as I was by this encounter with previously unseen, unknown networks of communication, we had to move on. After a couple of hours we got back on the bus, returned to the city and resumed our conversations about technology, the internet and the modern networks which seem to have supplanted, or at least drowned out, those of the forest.

It was another year, and a work of fiction that brought home to me the significance of what I'd learned in that brief encounter. In Richard Powers' novel *The Overstory*, a variety of characters find their lives increasingly intertwined with, and dependent upon, the trees around them. In one chapter, a mulberry tree planted by a recent Chinese immigrant to America in his new backyard becomes the frame for a saga of family life over decades: a proposal, a child, a death and more. In another, an artist discovers a box of photographs taken by his great-grandfather, grandfather and uncle over a hundred years. The photos are of a rare Midwestern chestnut on the family farm, snapped every year, from the same place: a century-long time lapse of a growing tree emerges as the world shifts around it. Gradually, mediated by their encounters with trees, the book's various characters band together to campaign for the preservation of some of the oldest and grandest trees on the continent: 2,000-year-old Californian redwoods, almost completely destroyed by logging.

One of these characters is a tree researcher named Patricia Westerford. While still in graduate school, she starts to think differently about the trees she studies, setting herself apart from the other students. 'She's sure, on no evidence whatsoever, that trees are social creatures. It's obvious to her: motionless things that grow in mass

mixed communities must have evolved ways to synchronize with one another. Nature knows few loner trees. But the belief leaves her marooned.' Her first paper on tree communication – an analysis of maples which send out signals to warn one another of insect attacks – is savaged by established critics. For years, she is marginalized and ridiculed by other scientists for her assertions.

For me, Westerford is the most compelling character in the book: someone who feels a truth deeply and who insists on doing the dogged work of science, and of literature, to bring that idea into a meaningful, communicable framework of knowledge. Despite being fired from her academic position, and enduring years of itinerancy, she sticks with the trees, disappearing into the forests and living in a network of remote cabins, before discovering she's been vindicated in her absence and another generation of students has picked up her work and run with it. 'Her public reputation, like Demeter's daughter, crawls back up from the underworld. A scattering of scientific papers vindicates her original work in airborne semaphores. Young researchers find supporting evidence, in species after species. Acacias alert other acacias to prowling giraffes. Willows, poplars, alders: all are caught warning each other of insect invasion across the open air.' Eventually, she writes a book, *The Secret Forest*, which becomes a bestseller.

In this case, art follows life. Powers based the fictional Westerford's writings and discoveries on Suzanne Simard's very real work in the forests of British Columbia. The revelations about the other lives of trees which enthralled me in *The Overstory* were recountings of the same stories I heard beneath the trees in Vancouver. I already knew these things. I just hadn't understood what they meant.

The achievement of *The Overstory* is to make these revelations meaningful at human scale, building the connections between us and the trees around us. Powers uses the scale of arboreal life – its extent in time, as well as its size – to tell a new kind of epic story: multi-generational, planetary-scale and ecological, in the sense of deeply intertwined with and responsible to our environment. As I read it, I felt something shift in myself, a sense of having been blind all my life to events and processes, whole other lives that surround us all the time. Having always been more interested in words on pages and code on screens, I suddenly found myself leaning out of windows to touch

the leaves of nearby trees, and stopping in the street to trace, with wonder, the whorls and cracks of living bark.

This came as something of a shock. I did not think I was inattentive to the weird other lives of things. As an artist, technologist and technology critic, I've worked extensively with the new techniques of artificial intelligence, which I first studied almost two decades ago, but which have undergone something of a renaissance in the last few years. Attempting to understand these technologies, I've built machine-learning systems that seek to question our assumptions about machine intelligence and human agency, and the path they seem to be taking us down. The self-driving car was one such experiment; in another, I used the latest machine-learning technologies to simulate a relationship between the British climate and Brexit voting patterns, drawing parallels between stormy weather, cloud computing and cloudy thinking. And what had struck me each time about these engagements with machine intelligence was their sheer strangeness, their fundamental difference to our own ways of seeing and thinking. And yet, I had not extended such awareness to the even stranger intelligences which, it turned out, were already all around me.

Powers and Simard suggested that another way of seeing the world was possible, one infinitely more vital and interconnected than any I had previously imagined. In their worlds, information pulsed beneath the ground and floated on the breeze, interactions pulsed and shifted to the rhythm of the seasons, and knowledge and understanding grew, slowly but sturdily, over decades and centuries. Beyond the human, beyond the animal, now appeared another domain – multiple kingdoms – of flourishing, active, even intelligent beings: plants and fungi.

I have not always been the best of friends to plants. I do not have green fingers, and many have died at my hands over the years. But I am lucky enough to live with someone with a very different aptitude, so that in the last few years I have become surrounded by them. First our apartment, then our garden, bloomed with vines and vegetables, succulents and flowering plants, a riot of ever-growing, ever-changing life. Presence and attention breed awareness, and as a result of this closeness, I have begun to distinguish species, shapes, colours and preferences in ways which were previously inscrutable to me.

We have already seen how the forms of bodies and their relationships to one another shape the subjectivity of animals, their *umwelt*. Think of the arboreal gibbons, directing their attention and their intelligence upwards into the trees; the way an elephant's abilities are framed by its trunk; or the complex group dynamics of baboons and macaques. Intelligence, it seems, is something physical and relational, not a wholly abstract process, but one closely tied to our being and doing. And if every organism has an *umwelt*, what would the *umwelt* of a plant look like? How do they sense and act, perceive and contribute to the world around them? While reading about the busy lives of trees in the forests, and the chatter passing through the mycelium, I looked around me and started to wonder what I might learn from these potted individuals, these trailing legumes and budding brassicas, cacti and lemon trees, and the way each individually responds and contributes to the world.

On the one hand, to get to know plants means to come down to earth, both literally and figuratively: you need to put your nose in the dirt, get mud beneath your fingernails, become familiar with the moistness of soil and its composition. The first effect of such familiarity is to remind us of the importance of place: specific places, with their own relationship to light and heat, sun and rain, shade and soil types. The abstract becomes specific. On the other hand, plants also connect us to everything else, again both literally and figuratively: their roots intertwine beneath the earth, the mycorrhizas grow together and their seeds scatter on the wind. They also precede us, and make life, everywhere, possible: through photosynthesis, soil production and the conversion of nutrients into food, plants are the founders and sustainers of the world.

It is these qualities which make plants good companions in thinking about technology. Both are endlessly generative, complex systems at both the macro and the micro scales. On the one hand, our contemporary technologies of power generation, information processing, communication and sensing are critical life-support mechanisms, which reproduce at a global scale and grant us – and themselves – near limitless, god-like powers. On the other hand, they depend upon the most minute of connections and the tiniest of electrical impulses; they need firm ground, the right atmosphere and constant human

attention. No wonder then that the poet Richard Brautigan was moved to imagine 'a cybernetic forest / filled with pines and electronics / where deer stroll peacefully / past computers / as if they were flowers / with spinning blossoms'.[2]

Somewhere between these two extremes – between the plant pot and the continental forest, between the microchip and the satellite – lies our actual lived experience, the place where our shared *umwelts* meet and mingle. For surely, as much as we differ, there must be points of overlap in our awareness, our sense of the world, which provide an opening to understanding. This turned out to be the case with my self-driving car; it was what Barbara Smuts discovered with the baboons. To understand a little of what plants mean and why they matter, we must discover what we have in common: the ways in which we share a world.

One way we might do this is to ask what plants hear. This simple question is already rife with contestation – plants have no ears or other known receptors for sound waves. And yet, quite assuredly, they hear, as we shall prove. How they do so, what they do with that information, and what it means for our interrelationships are similar questions to those we ask ourselves about animal intelligence: how are we changed by encountering non-human senses and a non-human impression of the world?

In 2014, two biologists at the University of Missouri recorded the sound of cabbage white caterpillars feeding on a cress plant. (*Arabidopsis thaliana*, rock cress, is the macaque of the botanical sciences, the most popular plant for biological experiments, and it has taught us many things about plant growth and genetics. It was the first flowering plant to have its genome sequenced and its DNA cloned, and it has even gone to the moon.[3]) Having left the caterpillars to munch away for some time, the scientists then removed them and played the sound of their approach back to the plants. Immediately, the plants flooded their leaves with chemical defences intended to ward off predators: they responded to the sound as they would to the actual caterpillars. They heard them coming. Crucially, they didn't respond in the same way when other sounds – of the wind or of different insects – were played to them. They were able to distinguish between the different sounds, and act appropriately.[4]

Claims about plants' capacity to hear and respond are not new. Neither are they always reliable. Notoriously, the bestselling 1973 work of pseudoscience *The Secret Life of Plants* used lie detectors and tape recorders to 'prove' that plants possessed emotions, telepathic abilities – and hearing. Roundly decried and debunked by scientists, the book's claims linger in the popular imagination, not least in the persistent belief that plants like classical music, or grow better when people talk to them. Fond as I am of all kinds of magical and imaginative thinking (and particularly, in this case, of Stevie Wonder's wonderfully odd concept album *Journey through the Secret Life of Plants*), the reality – as witnessed in the 2014 cress experiments – is better.

The cress displayed what scientists call 'an ecologically-relevant response to an ecologically-relevant stimulus'; that is, a plant-y response to a plant-y event. While plants are unlikely to have any preference for Beethoven or Mozart, for English or Arabic, the munching of caterpillars, on the other hand, has for plants a clear meaning – and the plants respond meaningfully in turn.

'What is the ecological relevance?' is the core question which rigorous science asks of any claims made about non-human abilities. In order to make sense of such abilities, it is necessary to frame them within an ecological context, the context of their relationships. It is not enough to say simply, 'this happened'. We must place each action within its context and understand why it occurs: in relation to which particular pressure, need or desire, or stimulus. Only then can we say that we truly understand an ability, rather than simply observe it.

This is a double-edged sword. On the one hand, it restricts our understanding of those abilities to an understanding of their circumstances, and as we already know from animal studies that can be woefully limiting. It's precisely what stopped us from recognizing the intelligence of gibbons, dolphins and others for so long, because we didn't fully understand their context and so missed the relevant cues. And whose understanding are we even talking about? The gardener, the greenkeeper and the forest-dweller have a very different perception of a plant's situation from that of the lab technician. On the other hand, the focus on ecological relevance pushes us to focus on relationships rather than innate abilities: on what matters when plants, animals, we and

others meet. What matters in these encounters and how do they shape our relationships?

If the context of our relationships really matters so much, then it will also matter in our relationships with machine intelligence. Perhaps we should be thinking more carefully about the ecosystem in which we are raising AI, particularly the kind of aggressive, domineering and destructive forms which seem to be proliferating. That these systems are overly concerned with profit and loss, winning and losing, control and dominance, suggests that their ecological niche – the slice of the environment shaping their evolution – is somewhat narrow. Their learned responses are that of a corporate intelligence, evolving within the arid, airless ecology of neoliberal capitalism, tech company boardrooms and ever-increasing financial and social disparities. If we wish them to evolve differently, we will need to address and alter this ecology.

The concept of ecological relevance admits something else too, something even more earth-shaking. It demonstrates that plants have a world. What does this mean? It means that plants sense and respond to a world which they experience, a world of their own making – and, furthermore, that there is a 'they' there to do the sensing and responding, a subject rather than an object, a kind of self, however abstruse and unlike our own. They have an *umwelt* all of their own. Plants encounter, access, influence and are influenced by the world on their own terms and in their own fashion. Most of these experiences and responses are and will always be unknowable to us; but hearing – being an ability we share – makes them thinkable. Suddenly, from being background, plants leap into action once again, present and attentive. The act of hearing transforms vegetal passivity into active listening, leafy torpidity into vibrant participation.

Of course, that's only our realization, our novel awareness. They've been active all along. Just as they probably don't care about Mozart or Stevie Wonder, so they don't need us to know that they are there. We choose to acknowledge them because it enriches us.

One morning some years ago, I was walking with my partner on the slopes of Ymittos, the mountain which rises on the eastern edge of the city of Athens. It was the day before spring – not the spring of the calendar, but the spring of the year, when everything bursts forth at

once on an unspoken cue. The whole mountain seemed to be preparing for it. As we descended through the forest around the ancient monastery of Kaisariani, we came across a cherry tree on the very threshold of flowering. Its blood-red bark was stretched tight around its trunk, each tiny leaf held in tension, each branch tipped with a swollen pink bud, its tip pursed in anticipation: the whole thrummed with a barely contained energy, as if straining against a tightly held breath. I'd never before experienced a being so insistent upon life without visible movement. I've since come to realize that this is how trees, plants and much of the earth are, much of the time. This feeling has become a part of me, and I am capable of calling it up at almost any time, simply by turning my conscious attention towards the whisper of the trees and the turning of the leaves. *We share a world.*

This shared world is not flat, nor singular. Many worlds – lively, noisy worlds – exist; many don't include us at all. Experiments with plant hearing suggest that, unsurprisingly, in many of those worlds the footsteps of caterpillars matter more than classical music or human language. It is a beautiful illustration of the kind of decentring of ourselves and of human experience at which we must become adept in order to live better and more responsibly in a more-than-human world.

This decentring, an admission that the human race is not the only game in town, does not correspond to any reduction of our world. Rather, as when we extend the virtue of intelligence to other beings, the addition of plant worlds to our own enriches both. Even those worlds in which we do not participate add to the totality of sensations and experiences which form the living, teeming earth, and on which we live and depend.

The very existence of other worlds, of numerous overlapping worlds in which many kinds of things and many ways of seeing and being are possible, should thrill us. Other worlds are not only possible, they are already present. The acknowledgement of multiple other worlds, the worlds of others, is key to disentangling ourselves from our greatest social and technological deception, and re-entangling ourselves with a more meaningful and compassionate cosmology.

I call this deception the 'one world' fallacy. Since the Enlightenment

of the seventeenth and eighteenth centuries, and the scientific revolutions that followed it, our understanding of the world has been shaped by a misplaced objectivity – the belief that the world has a single, coherent narrative and that there exists a one-size-fits-all framework for interpreting it. Eighteenth-century scientists called this 'truth-to-nature', although their attempts to fix an understanding of how nature 'really' worked involved eliminating its inconsistencies and idiosyncrasies. As scientific practice has matured, we've come to understand the value of interpretation, situated knowledge and trained judgement; but this fallacy has been perpetuated by our technologies, which flatten and lump together the myriad different expressions of the world.

The power of these tools to make the world legible to humans in turn contribute to techno-determinism and network power: the belief that the tools produced in the most dominant societies by the most dominant groups are not only the best for everyone, but are inevitable and unassailable. And so, in contrast to the original intent of scientific investigation, a mismatch arises between the world as we perceive it, and the world as it actually is. The attempt to force the world to conform to our portrayal of it, and the friction this attempt generates in our lives and societies, is behind the great malaise of our age: widespread confusion, shading into anger, rage and fear. It is the result of trying to find truth and meaning in a single world, a single box into which we cram all the contradictions and paradoxes of reality. But in truth, there are so many worlds.

The fact that we are still able to live, to function, to survive and thrive together in this world-of-many-worlds, also implies that these worlds are shared. There are points and planes of intersection, shared experience and shared awareness. All the inhabitants of the earth – animals, plants and diverse others – are, whether they care about classical music or not, whether we even notice it or not, buffeted by the same vibrations in the atmosphere. By dispensing with the fallacy of one world for all, we come to the awareness of a greater multiplicity of worlds which are held in common. This is a far richer cosmology than the solipsism of one world; it is an acknowledgement of communal being and experience. *We share a world*. We hear, plants hear; we all hear together. We all feel the same sun, breathe the same

air, drink the same water. Whether we hear the same sounds in the same way, whether they are meaningful to us in the same way, is beside the point. We exist, together, in the shared experience and creation of the more-than-human world.

This chain of realizations is significant, it seems to me, for it fundamentally transforms our possibilities for thinking not only about the lives of plants and trees, but about all these other minds, beings and persons. It permits a wholly different kind of thinking, one that is more open and clear. Because acknowledging the existence of non-human worlds, and subsequently the existence of a shared world, helps us navigate the twin hazards that we face in thinking about the more-than-human world: anthropocentrism and anthropomorphism. The former is the danger of thinking ourselves to be at the centre of everything; the latter is the danger that, in trying to access non-human experience, we simply mould it into a poor shadow of our own.

Fully recognizing that non-human plants, animals and others have their own worlds which are fundamentally different and unknowable to us is to begin to end human exceptionalism and human supremacism. Humans are not at the centre of the universe. By stepping aside we can start to imagine what a world in which we are not the most important thing might actually look like, and consider the richness of non-human worlds on their own terms.

Yet some measure of anthropomorphism – the attribution of human traits, emotions and intentions to non-humans – is inevitable. It is, when all's said and done, the only way we can begin to understand other living things. We are *anthropos*, and we have no other means of addressing these worlds than through our own. But if we can consciously bear in mind the fact of our difference, if we can recognize our own limited perspective while not enforcing it upon others, these things need not be a barrier to accessing and acting on the basis of shared interests and intentions. In fact, it is precisely through the acknowledgement of difference that shared action and shared life, without the weight of domination and control, becomes possible. We seek to meet, and not to conquer. We are as comfortable with companionable silence as with talking. We have different words for the same things.

This position even allows us to think again about abilities that are *not* ecologically relevant, or not apparently so. We still have no idea

why bird eggs are brightly coloured, for example (seemingly an invitation to predators), or why plankton are so diverse, defying the laws of evolutionary competition.[5] But we don't need to have a complete explanation for every mechanism before we test our theories, because – contrary to the design of most scientific experiments – we live in the same world as our test subjects. This is brilliant: suddenly whole vistas of inexplicable but obvious behaviours become interesting and thinkable.

One such extraordinary behaviour is the ability of plants to remember. That plants hear, in some form, is not too great a stretch for our imaginations. But the idea that they might remember – and therefore perhaps think and reflect, and do all the things we humans do with memory – well, that seems like a stretch. After all, we have no inkling of the structures and processes which might facilitate such behaviour. And yet, we do know that plants remember.

The greatest – and most controversial – exponent of plant memory is the Australian biologist Monica Gagliano, whose scientific background is an object lesson in ecological thinking. Gagliano began her academic career as a marine ecologist in the 1990s, studying the behaviour of fish on Australia's Great Barrier Reef, before turning to the plant world. It's this background which informs her working practice. Historically, scientists' approach to the study of plants has been a mechanistic one; that is, they break them down into a series of actions and reactions, viewing them more like a series of component mechanisms, as a series of tiny, interconnected machines, rather than as whole organisms. And this means we regard them in much the same way as we regard machines, capable only of reacting to stimuli in automatic, pre-determined and predictable ways. Botany sees only narrow cause and effect, and often fails to fully consider the whole life and experience of the organism. In contrast, Gagliano sees plants as having behaviours, like the tropical reef fish she studied previously.

Gagliano's favourite partner in this research is *Mimosa pudica*, also known as the 'touch-me-not' or 'sensitive plant'. The mimosa displays a particularly startling behaviour: on being touched, with a stick or a hand, it suddenly and quickly rolls up its leaves into a tight bunch. It belongs to a rare class of plants whose movement is visible to the naked

eye: only the Venus flytrap, with its snapping jaws, and the Asian telegraph plant, whose dancing leaves entranced Charles Darwin, can rival it for speed. As such it makes an excellent and communicative partner in experiments.

Gagliano wanted to change scientists' understanding of plants, to show that they were capable of learning from experience, and altering their behaviour as a result, an ability scientists had tended to reserve for 'higher' animals.[6] In order to demonstrate this, she devised a simple mechanism: a cup attached to a vertical rail which, when released, would drop a mimosa in its pot exactly fifteen centimetres onto a foam pad. The soft *thonk* of this landing was enough to surprise the plant into closing up its leaves, but not enough to damage it: a scientifically precise stimulus, with a measurable, ecologically relevant response. Mimosa leaves have small hydraulic structures at their base which, by pumping or draining water, allow them to expand and contract, forcing the leaves to curl. The curling is understood to be a response to a threat: either from animal predation, or excessive heat and evaporation. Either way, a traditional cause-and-effect study would probably stop here: a relationship between stimulus and response has been established, the mechanism known and understood, our expectations of plant capabilities satisfied but hardly enlarged.

Gagliano went further. She kept dropping the plants, over and over again, up to sixty times a session, over multiple sessions. Three hundred and sixty drops a day: a marathon of bumps and shocks. *Thonk thonk thonk*. It transpired that it only took a few drops – as few as four or five – for the plants to realize that there was no threat, and that it was safe to keep their leaves open (a side test, involving a different stimulus which still elicited the closing response, revealed that they weren't simply exhausted by the activity). By the end of the sixty-drop sessions, the plants were entirely unbothered by the drop: they had learned to ignore it.

Understanding a lesson is one thing; its value depends on the ability to recall it later, to put the knowledge gained into practice. So Gagliano and her colleagues rested and then retested individual mimosas, demonstrating that they retained over time the memory of the drop, and their associated change in behaviour. Mimosas – and, we must now understand, all plants – are not machines. They are more than

the sum of a set of pre-programmed actions and reactions. They learn, remember and change their behaviour in response to the world.

This is an astonishing discovery. Not only does it run counter to everything we thought we knew about plants, but it radically overhauls the very categories of knowledge we have for understanding them. So much so that it took years for Gagliano to publish her results, following rejections by over a dozen academic journals, many of which would not even acknowledge reading her unpublished paper, let alone send it out for peer review. Part of the problem is innate academic conservatism – upending decades of understanding is not only hard to do, it's hard to imagine and acknowledge – but another part is simply bureaucratic. Where to put this new knowledge? Whose domain does it fall under? Is this botany or physiology, ethology or psychology?

As Gagliano put it, 'We were somewhat unrehearsed for the play *Mimosa* had put on and which – candidly unprepared or stubbornly unwilling – we were all, nevertheless, characters in.' The fact of *Mimosa*'s behaviour was undeniable, and it demanded we rethink our assumptions. Science has a tendency – a prime directive, even – to understand phenomena through the structures that produce them; that old ecological relevance again. It understands memory as a structural process which takes place in the brain, in the interaction between particular molecules and synaptic connections. In this sense, memory depends on the brain. How then can something without a discernible brain remember? It's simply unthinkable.

But we can look at it another way. Rather than denying the existence of plant memory because plants lack brains, discoveries such as Gagliano's give us the opportunity to rethink what memory is, and admit that perhaps it can arise by other means, in other structures, in other beings. Like the intelligence of the octopus, the memory of a plant is an analogous phenomenon running on completely different hardware.

One intriguing aspect of Gagliano's account of her research – and one which puts many other scientists' backs up – is that, for her, each experimental result is the result of a plant speaking. She means this literally: as a result of a long-standing practice of meditation and shamanic rituals, she has received direct instructions from entities she

considers to be plant spirits, who offer her advice on how to conceive and structure her experiments. Each of these encounters is then an introduction, an opening of the channel, which allows the experimental subject to speak for itself, for *Mimosa* to declare its ability, now that we are ready to hear it. And every time the plant speaks, the boundaries which we thought enclosed our worlds are shaken.

If you're sceptical, then fine. But it's not necessary to accept the reality of Gagliano's direct communication with plant spirits to accept the results of her experiments, which in every other way conform to the most stringent scientific norms, including peer review and reproducibility.[7] I find aspects of Gagliano's account hard to take seriously too. Yet I've had my own personal experience with Ayahuasca, one of the plants she speaks with, and it has spoken to me, too. I have heard a plant speak, and I still don't fully understand how, nor can I adequately describe the experience, but I know it to have happened, and it changed me utterly.

Western scientific conditioning is hard to break with. It makes it hard for us to trust even our own experiences, whether that is the ecstatic experience of spirit encounters in the grip of psychedelics, or the uncanny sensation of communion in the eyes of other species. This is part of the genius of Gagliano's experiments: the rigorous application of the scientific method, the careful weighting and testing of her *Mimosa* plants, and the reproducibility of her method means we don't have to put all our trust in our subjective experience. What is required of us is to be open to changing our minds.

Mimosa speaks, and the world changes. We are forced to account for a different reality than the one we knew. When we open ourselves to the voices of the more-than-human world, this is always the result: a breaching and collapsing of the established borders of thought and feeling. This is the outcome that shines through these encounters, with all kinds of intelligences, including 'artificial' ones. Machines beat us at chess, take the wheel of the car and make scientific discoveries, and what we thought were uniquely human endeavours become instead shared ones. Baboons stare deeply into pools of water, and the experience of contemplation we thought wholly our own is suddenly mirrored in, and enhanced by, that of others. Plants have a world, it's different to our world – and, oh, we share a world! Each time, acknowledgement of

the agency and difference of the other actually brings us closer, while broadening our own vision, knowledge and experience. It enlarges us.

The field just grows and grows. Plants don't just hear and remember – they make their own sounds. In another experiment, Gagliano recorded the sound of corn kernels clicking at frequencies far beyond human perception, their purpose still unknown. It turns out plants can smell too, detecting both predators and the warning signals given off by neighbouring plants. They even smell out prey, as animals do: the parasitic dodder vine sniffs out suitable victims, which it then envelops and feeds upon. Plants make decisions based on complex information, such as picking the best response to nearby competition. Grow sideways, root deeper or shoot higher? They can tailor chemical releases to attract or repel animals – poisoning some and creating addiction in others. They have proprioception – the 'sixth sense' which enables us to know where parts of our bodies are without looking. They recognize and respond differently to kin and close relatives. In short, plants act, and they act in ways which, when animals do likewise, we call indicators of intelligence.[8]

In just the last decade, we have crossed a once-invisible line in our relationship with plants: they have been transformed in our understanding from objects into subjects and, as in any conversation, their subjecthood will continue to expand in scope the longer we converse. As with the abilities of animals, this tally of plant agency will grow and grow as we admit more and more possibilities, come up with new theories and design the apparatus to test them.

This is not to make any specific claim about the intelligence of plants. The exact form of plant intelligence must always remain partially or mostly unknowable to us, because of the radical difference which exists between our own lives and our experience of the world, and that of plants. But because we share a world, we can think about what intelligence and memory might be, when we include others in that thinking, when we understand it as something which acts between us and in the world, rather than merely in our own heads. You can read endless books about plant and animal intelligence but they'll all tell you the same thing, that non-humans are brilliant and also unknowable, and that the greatest joy in the world is to be found not in testing and taxonomizing but in going on together.

How we talk about this newfound subjecthood of plants is another matter. How do we make sense of these newly recognized skills, this life that no longer merely teems, but thinks? It's tempting, as with non-human animals, to speak immediately of plant intelligence as if it were comparable to the human kind, to induct the geranium and the vine into that exclusive club that we're just starting to shake up. But, again, that seems to miss the point. Like the restriction of 'artificial' intelligence to the kind of things that humans do, it runs straight back into anthropomorphism and anthropocentrism. Whatever plants are doing, they may resemble us and other animals in some ways – but they are clearly different in other, significant ways. The insufficiency of intelligence as a way of explaining these abilities is clear. Plant intelligence, whatever it is, is plant-y.

Debates over plant intelligence, and animal intelligence, and artificial intelligence, and even human intelligence, are thus revealed as endless – and also, ultimately, pointless, because they are debates about the meaning of words rather than the being of things. Where we start to move forward is when we learn to ask questions which are less concerned with 'Are you like us?', and more interested in 'What is it like to be you?'

The reason we have so much difficulty understanding what plant intelligence might consist of is precisely because it is so different to our own. One obvious objection to the idea of plant intelligence is that plants don't have brains. Yet one of the strengths of plants is precisely that they have no central, irreplaceable organs. Plants are modular – they can survive losing 90 per cent of themselves, and many species can reproduce from broken pieces, or cuttings. In particularly harsh environments, like deserts, cuttings are more successful than seeds for raising new generations, which is why many succulents are particularly good at this kind of reproduction. Modularity creates resilience: fetishizing particular organs, as animals do, obscures such abilities. Rethinking what intelligence might be also allows us to rethink the modes and mechanisms which might produce it, and thus to come up with new ways of being intelligent.

Plants such as succulents also produce clones: genetically identical offspring which may separate from the parent, or remain attached, shaping their parents into new forms. While cloning can be a weakness,

as genetically identical offspring are vulnerable to the same threats as the parent, it can also create extraordinary creatures. One such clonal wonder is Pando, an aspen living in Fishlake National Forest in Utah. At ground level, Pando looks like a forest. They take the form of more than a hundred acres of quaking aspen trees; 47,000 tall, slender trees with white bark and black knots, whose leaves turn shades of brilliant yellow in the autumn. But in fact Pando is one individual, a single organism, in which each tree trunk is a shoot from a single root system. They are one of the largest and oldest individuals on Earth.

Pando was born around 80,000 years ago, or perhaps many, many more: one study puts their age at closer to a million years.[9] Clonal trees can't be aged by their rings, as each is a mere passing stem of the more ancient root. For most of Pando's life, conditions were close to perfect for a clonal aspen: frequent forest fires prevented competition from conifers, while a gradual climatic shift from wet and humid weather to drier, semi-arid conditions made it harder for younger aspens to seed and establish rivalries. In fact, hardly any new quaking aspen have grown from seed in the western United States for 10,000 years, yet the aspen remains North America's most widely distributed tree.

Today, Pando is believed to be dying – albeit very, very slowly. No new growth has been detected for several decades. The possible causes of this are numerous, but all point back to human interference. The establishment of cabins and campsites within Pando's expanse reduces the likelihood of the cleansing forest fires on which aspen depend, while the disappearance of bears, wolves and mountain lions has led to an explosion of mule deer. The deer, along with cattle, graze on young aspen shoots. Once areas are eaten or cut back, they don't regrow.[10]

Pando wasn't 'discovered' until the 1960s, not least because they don't look like a tree – they break with our idea of what a tree is. Yet as our idea of what a tree can be expands, so do our models for thinking about other structures in the world. Indeed, the discovery and understanding of plant communication itself owes much to such radical rethinkings of the way we see the world, and leads to new realizations in turn.

Earlier, I mentioned the mycorrhiza which underpin life in the forest: the networks which process, store, extract and transport

food and information between the surface plants. To think of these networks – of fungus itself – as separate from the plants which they connect is not really correct. As we've discovered in recent decades – like everything we've discovered in recent decades – the kingdom of fungi and the kingdom of plants are not really that separate. They are physically, socially, vitally entangled with one another. Understanding the nature of this interrelationship is vital to understanding our own entanglement with the more-than-human world.

The earliest plants were mere agglomerations of cellular tissue, lacking roots or leaves or any of the specialized structures we recognize in plants today. They were the descendants of simple marine algae which washed ashore and found some purchase on beaches and cliffs, sustaining themselves through photosynthesis alone. But around 400 million years ago – at least, that is the date of the oldest fossils we've found – these proto-plants began to associate with fungi: to evolve lobes and fleshy organs to house mycorrhizal partners. This is the origin of all plant roots: questing limbs in search, not of food itself, but of partners in the process of producing life.

Plants and fungi don't merely interact underground, they penetrate one another. Parts of fungi actually live within the cells of plants, and they form in effect an extended root system, more than a hundred times longer than the roots of the plant itself. And these fungal strands, called mycelium, extend everywhere and through everything. What we take to be the soil itself is actually part fungus – somewhere between a third and a half of its living mass. Plant, fungus, and the entire ecosystem on which we and all life on earth depends, is inseparable, right down to the cellular level.

This relationship between fungus and plant is symbiotic: each depends upon the other. As the mycologist Merlin Sheldrake puts it: 'What we call "plants" are in fact fungi which have evolved to farm algae, and algae that have evolved to farm fungi.'[11] We must then understand that our own lives are dependent, symbiotically, on such relationships: for the food we eat, the environment we inhabit, and even the climate we – hopefully, for now – enjoy.